With that, we have explored how to generate structured data using AI models and integrate it into our React components. As we move forward, in the last section of this chapter, we look at an even more advanced capability of AI models: tool and function calling. This feature takes AI interactions to the next level, allowing models to not just generate text or structured data but invoke specific functions or use tools as part of their response process. This allows AI models to perform actions, retrieve real-time information, or execute complex operations as part of their reasoning process.

4.4 Tool and function calling with AI models

AI models are effective tools that excel at generating text and media content based on specific prompts or queries. While these models possess vast knowledge, they may sometimes lack access to real-time, specialized, or rapidly changing information.

What if we could ask the AI model to automate parts of the investigation process to call tools that we provide to aid its results? For example, when asking what the weather is like in a particular region, the AI model would use an API that we provide, call this, and then respond with a structured format.

To address this challenge, developers have introduced tool-calling and function-calling capabilities, which significantly enhance AI models by allowing them to interact with external resources. At their core, *tool calling* and *function calling* in AI models refer to the model's ability to recognize when external information or actions are needed, request the execution of specific functions or tools, and incorporate the results into its response. This process allows the AI to obtain real-time or specialized data, perform actions beyond its training scope, and provide more accurate, relevant responses.

The basic idea of tool and function calling in AI models enables interaction with external functions or tools to perform specific tasks or retrieve information. In this section, we build on the basics we've learned so far in the previous sections about generating structured data and streaming content to instruct the AI model to use tools that we provide. This will unlock new important functionalities that we can include in our Astra AI agent.

4.4.1 Understanding tool calling and function calling in AI models

The process of tool and function calling typically follows a specific flow. First, the AI model identifies the need for external information or action based on the user's query or the context of the conversation. It then generates a structured request for a specific tool or function. The system receives this request and executes the requested tool or function, which might involve calling an API, querying a database, or performing a calculation. The result of this operation is then fed back to the AI model, which incorporates this new information into its final response to the user. Figure 4.4 provides a visual insight into how this process works.

In this process, the user's prompt is sent from their device to your application server, which forwards it to an external AI model service. When the AI model recognizes the need for a tool, it sends a structured request back to your server. This is where the